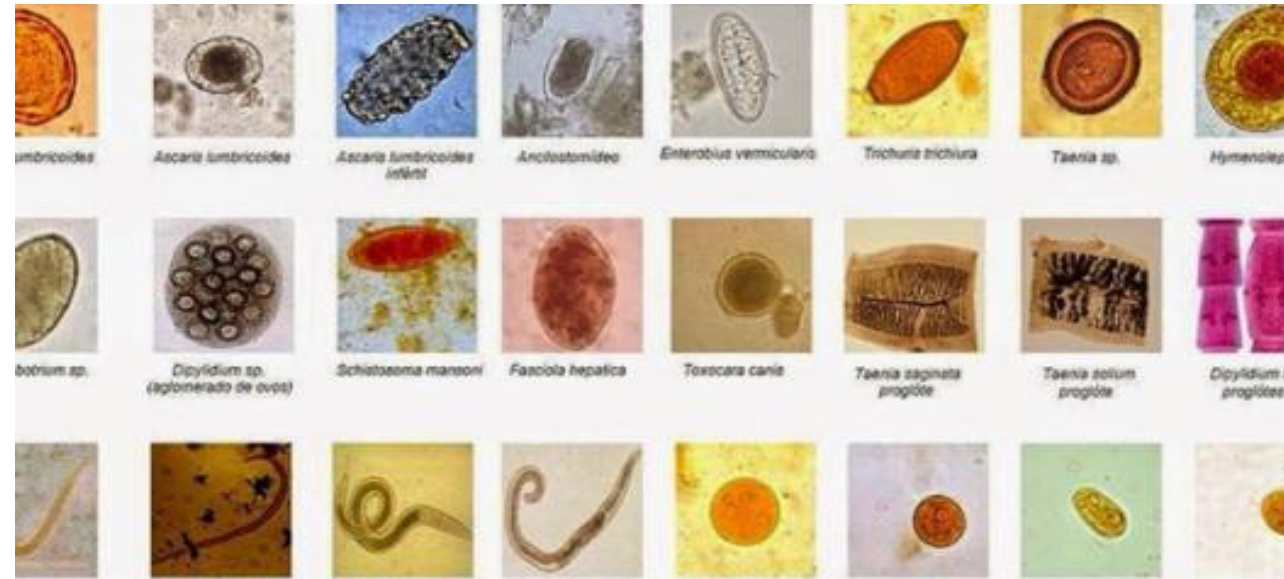


Introduction to parasitology



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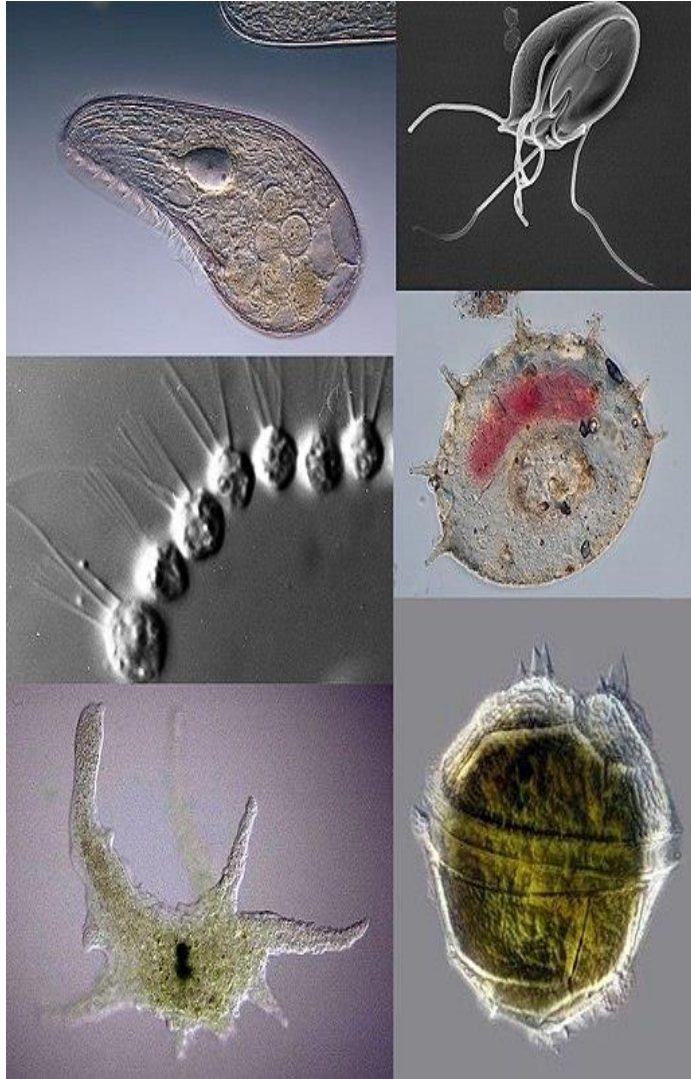
Medical parasitology

- The study of parasites is called Parasitology
- Medical parasitology deals with the parasites, which cause human infections and the diseases they produce.
- Medical parasitology is focused on medical parasites and includes their morphology, life cycle, and the relationship with host and environment.

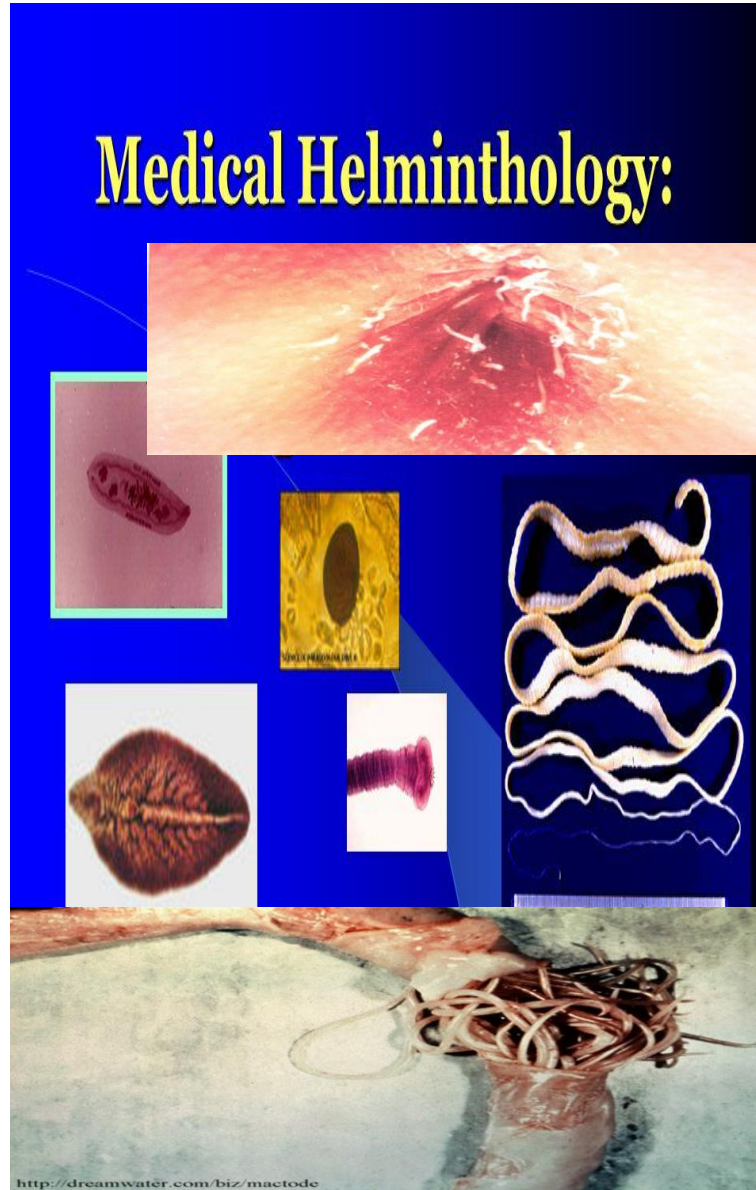
Medical parasitology is broadly divided into 3 parts:

- **Medical Protozoology:** deals with the study of medically important protozoa.
- **Medical Helminthology:** deals with the study of helminths (worms) that affect human
- **Medical Entomology:** deals with the study of arthropods that are ectoparasites and vectors which cause or transmit diseases to human.

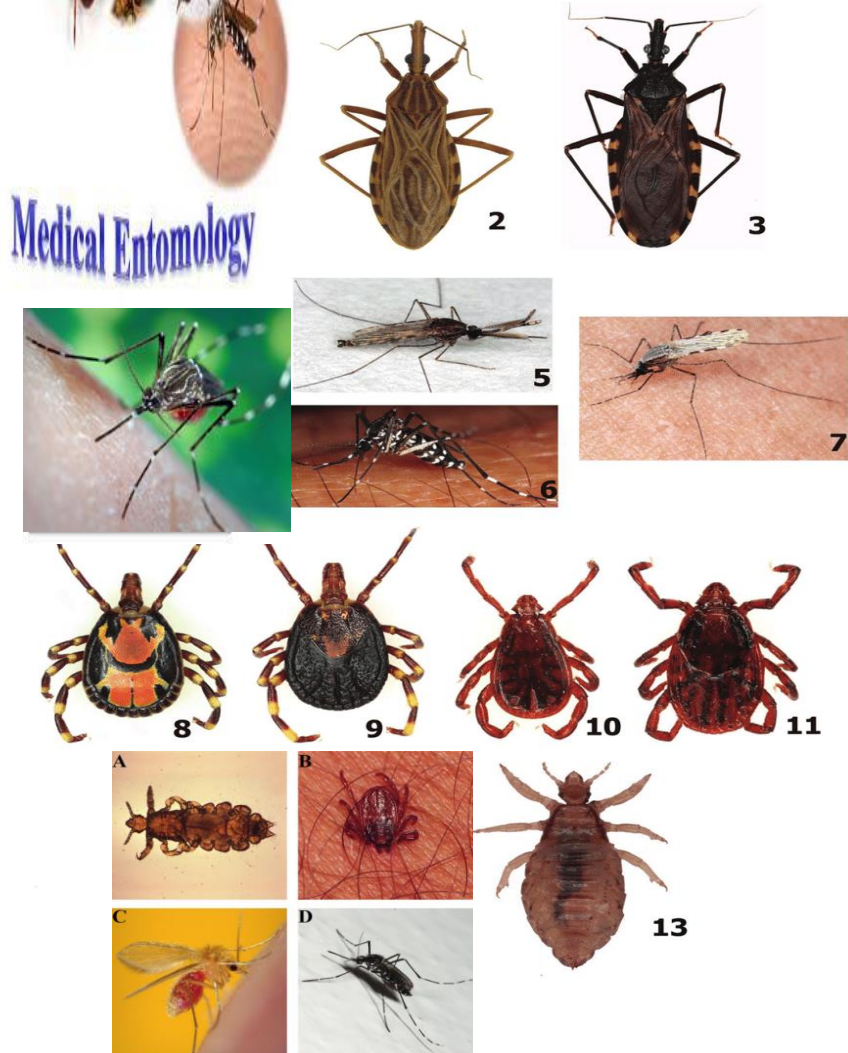
Medical Protozoology



Medical Helminthology:



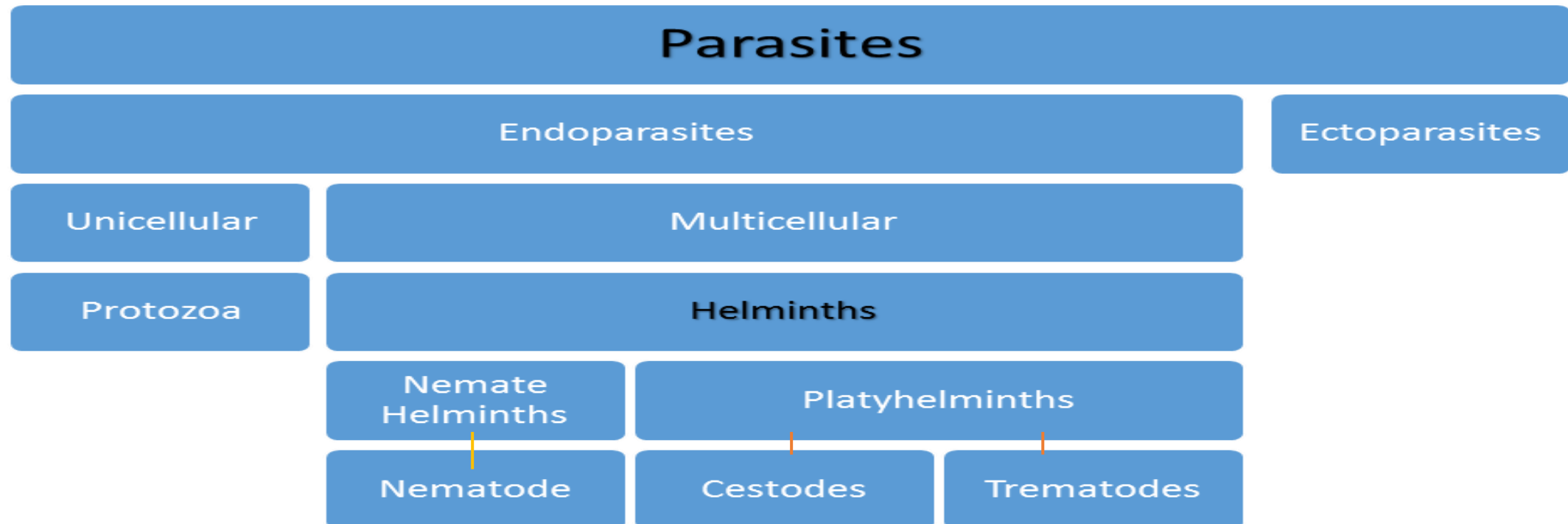
Medical Entomology



Parasites

Parasites: are living organisms, which depend on a living host for their nourishment and survival. They multiply or undergo development in the host.

Host: is defined as an organism, which harbors the parasite and provides nourishment and shelter to it and is relatively larger than the parasite.



The word **parasite** comes originally from the Greek *parasitos* (*para*, alongside of; and *sitos*, food), meaning someone who eats at another's table or lives at another's expense.

Classification of Parasites

- Kingdom
 - Phylum
- Class
 - Order
- »Family
 - Genus
 - Species



Scientific Nomenclature

Binomial System of Nomenclature:

- Each parasite belongs to: **Phylum, Class, Order, Family, Genus and Species.**
- Each parasite is written by Generic name (with 1st letter capital) and Specific name (1st letter small); and printed in italics or underlined.

Example: ***Schistosoma mansoni***

or **Schistosoma mansoni**

Genus

Species

Classification of parasites based on location

- Ectoparasite:

Ectoparasites inhabit only the body surface of the host without penetrating the tissue. Lice, ticks, fleas and mites are examples of ectoparasites. The term infestation is often employed for parasitization with ectoparasites.



- Endoparasite:

A parasite, which lives within the body of the host and is said to cause an infection. Most of the protozoan and helminthic parasites causing human disease are endoparasites. Examples (*Taenia saginata*, *Taenia solium*, *Ascaris lumbricoides*).



- Free-living parasite:

It refers to nonparasitic stages of active existence, which live independent of the host, e.g. cystic stage of *Acanthamoeba spp.*

Classification of parasites based on Habitat:

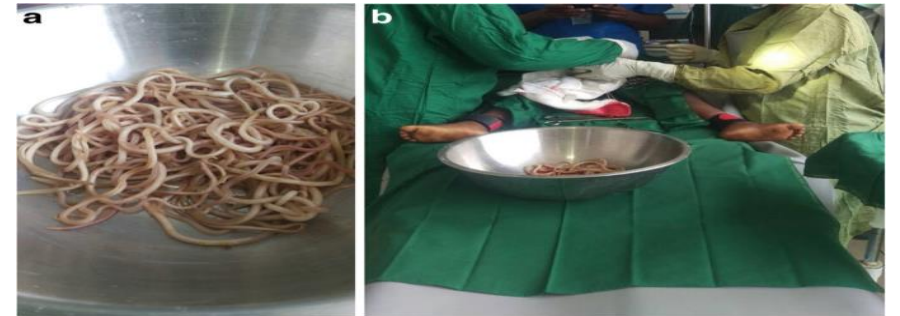
- **Obligate parasite:** The parasite, which cannot exist without a host, e.g. *Toxoplasma gondii* and *Plasmodium*.
- **Facultative parasite:** Organism which may either live as parasitic form or as free living form.
- **Accidental parasites:** Parasites, which infect an unusual host are known as accidental parasites (*Echinococcus granulosus* infects human accidentally, giving rise to hydatid cysts).
- **Aberrant parasites:** Parasites, which infect a host where they cannot develop further are known as aberrant or wandering parasites, e.g. *Entamoeba histolytica* in the liver or lung of humans.

Classification of parasites based on duration

- **Temporary parasites** live in contact with the host for only a part of their cycle or occasionally at the time of feeding, for example, mosquito, (visit a host for a short periods).



- **Permanent parasites** live in contact with the host throughout their life, for example, *Ascaris lumbricoides*, *Entamoeba*...etc



Host

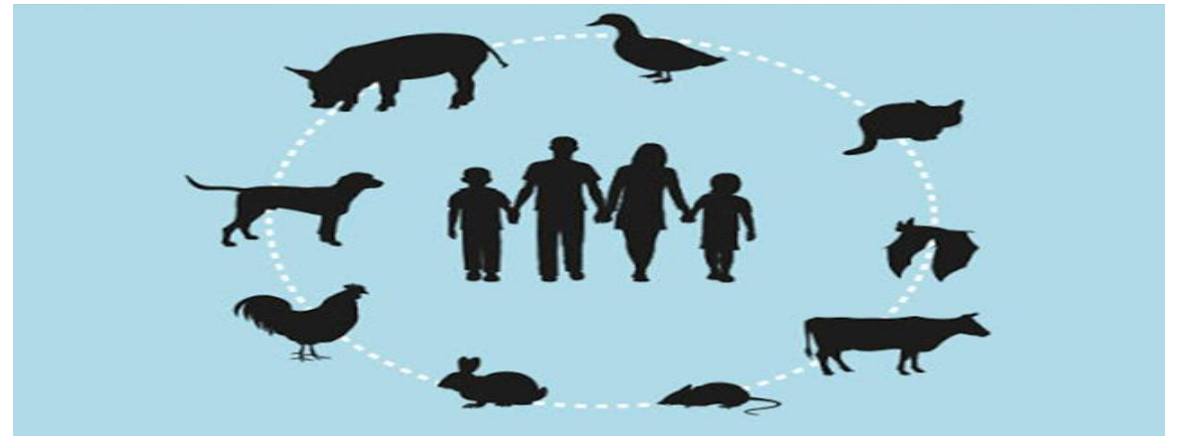
- The host may be of the following types:
- **Definitive host**: The host, in which the adult parasite lives and undergoes sexual reproduction, e.g. **mosquito** acts as definitive host in **malaria**.
- **Intermediate host**: The host, in which the larval stage of the parasite lives or asexual multiplication takes place. In some parasites, **2 different intermediate hosts** may be required to complete different larval stages. These are known as first and second intermediate hosts, respectively (snails are the 1st intermediate hosts & fish are the second intermediate hosts for (*Clonorchis sinensis*)).

- **Reservoir host:** a host that makes the parasite available for the transmission to another host and is usually not affected by the infection (**pig** is the reservoir host for *Balantidium coli*).
- **Natural host:** A host that is naturally infected with certain species of parasite (**canines** in *Echinococcus granulosus*).
- **Accidental host:** The host, in which the parasite is not usually found, e.g. **human** is an accidental host for **cystic echinococcosis**.
- **Carrier:** A person who is infected with the parasite but without any clinical sign and symptoms, and is an important source of infection.

- **Vector:** A vector is an agent, usually an arthropod that transmits an infection from man to man or from other animals to man, e.g. **female Anopheles mosquito** is the vector of malarial parasite.



- **Zoonosis:** refers to diseases which can be transmitted from animal to human, e.g. Toxoplasmosis, leishmaniasis, Hydatid disease, and Taeniasis.

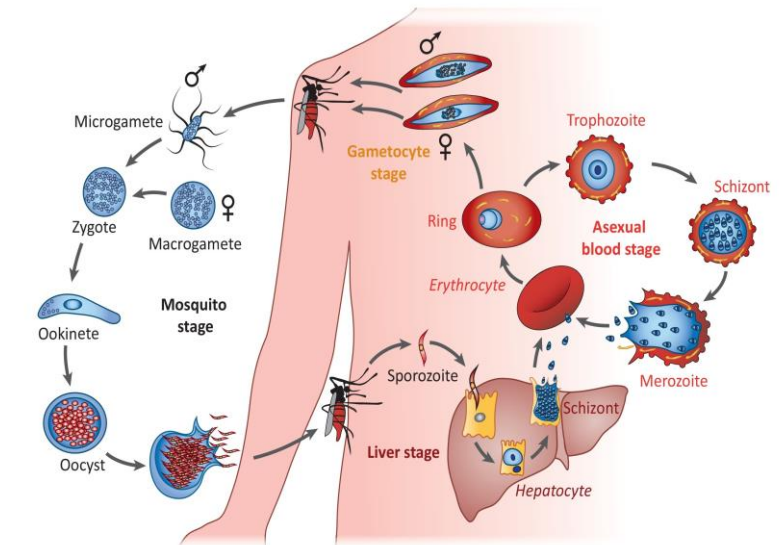
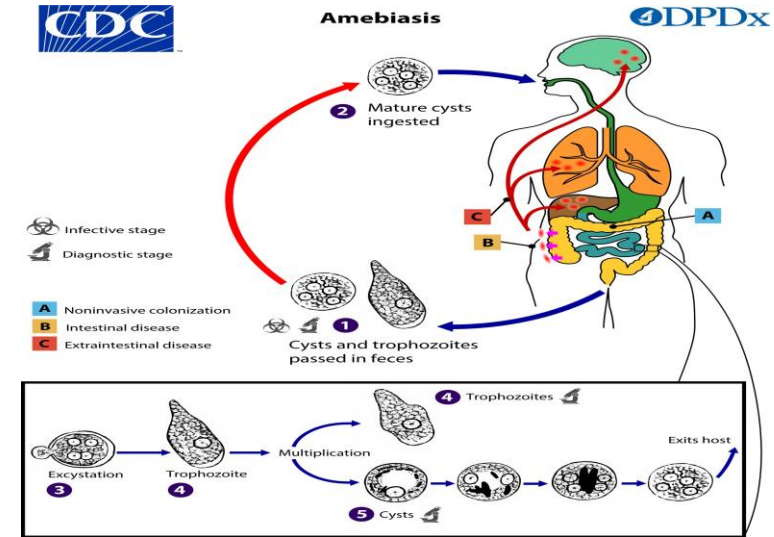


Host-parasite Relationships

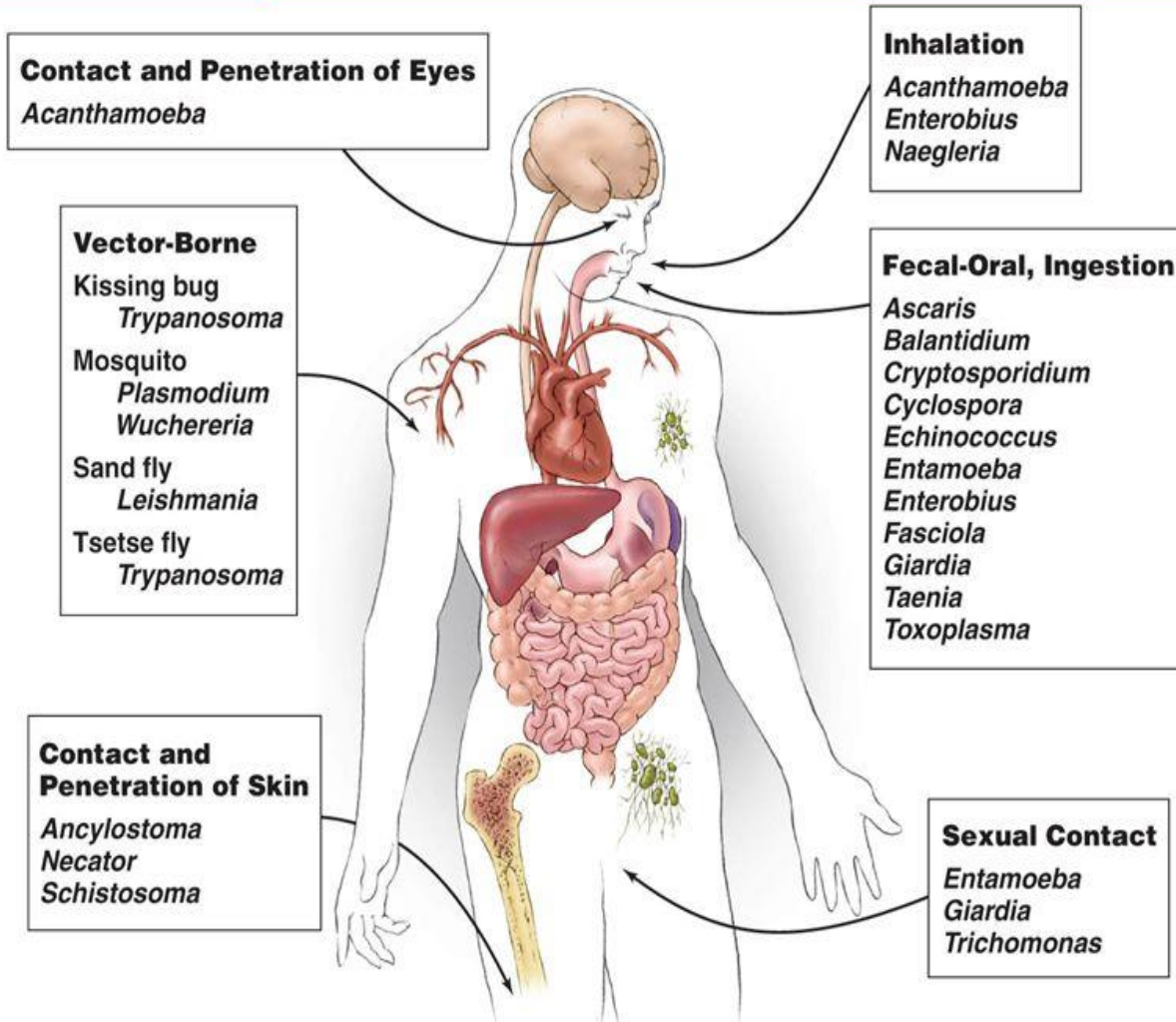
- Host-parasite relationships are of following types:
- **Symbiosis**: permanent association between two organisms.
- **1-Mutualism** - an association in which both partners are metabolically dependent upon each other and one cannot live without the help of the other; however, none of the partners suffers any harm from the association.
- **2-Commensalism** - an association in which the commensal takes the benefit without causing injury to the host.
- **3-Parasitism** - an association where one of the partners is harmed and the other lives at the expense of the other e.g. worms like *Ascaris lumbricoides* reside in the gastrointestinal tract of man, and feed on important items of intestinal food causing various illnesses.

Life Cycle of Parasites

- Direct life cycle: when a parasite requires only single host to complete its development, it is called as direct life cycle, e.g. *Entamoeba histolytica* requires only a human host to complete its life cycle.
- Indirect life cycle: when a parasite requires 2 or more species of host to complete its development, the life cycle is called as indirect life cycle, e.g. malarial parasite requires both human host and mosquito to complete its life cycle.



Routes by which humans acquire parasitic infections



- Oral transmission
- Skin transmission
- Vector transmission
- Direct transmission
- Vertical transmission
- Latrogenic transmission

Laboratory Diagnosis

On the basis of clinical features and physical examination laboratory diagnosis depends upon:

- Microscopy
- Culture
- Serological test
- Skin test
- Molecular method
- Animal inoculation
- Imaging
- Hematology.

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